KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 26 · MATH

The Motor City Robot Rally

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Scoring the Robots

Kavya's robotics team is competing at a robot rally in Detroit, Michigan. Warm up before the lesson, and show your work. Some problems may feel new, and that is okay!

1. Review (Unit 5): a robot's wheel guard is a regular hexagon with sides of 6 centimeters. a) What is its perimeter? b) How many lines of symmetry does it have?

2. Kavya's robot places 3 cones worth 4 points each. Write a multiplication sentence for the cone points, and find the total.

3. Kamau says 2 + 3 × 4 = 20. Kendrick says it equals 14. Who do you think is right? Explain your thinking.

4. What do you think the parentheses tell you to do first in (5 + 5) × 3? Then find the value.

5. Which expression matches "add 6 and 2, then multiply by 5"?

   (6 + 2) × 5 | 6 + 2 × 5 | 6 × 2 + 5

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: Judge the Scores

Grouping symbols first, then multiply and divide from left to right, then add and subtract from left to right. Show every step.

1 Evaluate each expression. a) 40 − 6 × 5; b) 72 ÷ 8 × 3; c) 2 × [18 − (5 + 4)]

2 Write an expression for each, but do not evaluate it. a) subtract 8 from 50, then divide by 6; b) four times as large as 327 + 85

3 Two steps: a robot places 9 cones worth 3 points each and 5 cubes worth 4 points each, then loses 6 points for a penalty. Write one expression for the score, then evaluate it.

4 Find the mistake: Kendrick says 36 − 12 + 4 = 20 because you add before you subtract. In two or three complete sentences, explain his mistake and give the correct value.

★ **Challenge:** Add one pair of parentheses to make each equation true. a) 3 + 5 × 4 − 2 = 30; b) 48 ÷ 4 + 2 × 3 = 24

Looking Ahead: Week 27: Powers of 10 and Exponents. At the Adler Planetarium in Chicago, you will multiply and divide by powers of 10 and write 10 × 10 × 10 as 10^3.